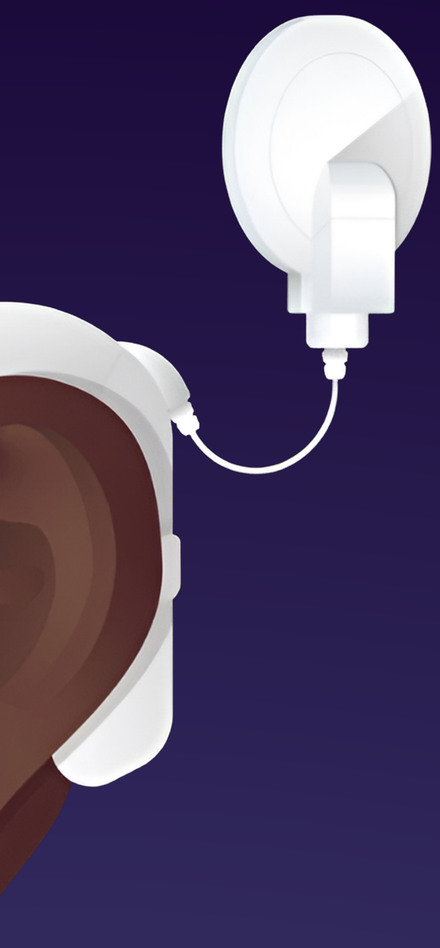


SOUND
Sensory Optimisation Using Neuroscience for Devices

UNIVERSITY OF CAMBRIDGE

Cambridge Hearing Group

HOW FREQUENCY-BASED PROCESSING MAY HELP CLINICIANS IMPROVE SOUND QUALITY PERCEPTION IN COCHLEAR IMPLANT USERS

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University Of Cambridge - Cambridge (United Kingdom)

INDIVIDUAL FACTORS

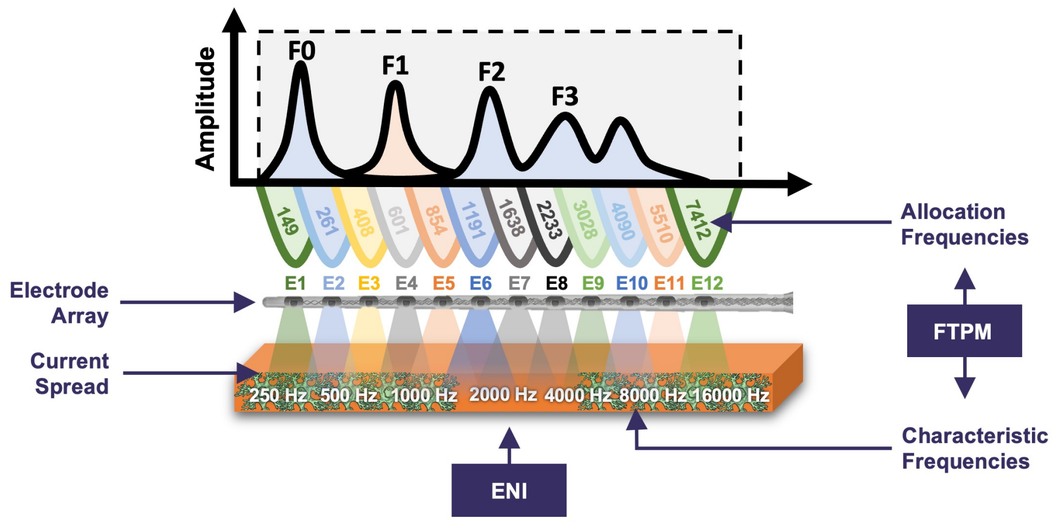
FREQUENCY-TO-PLACE FACTORS

FTPM[1] | ENI[2]

SOUND QUALITY

GOAL 1 Evaluate if different patterns of responses are obtained with the different participants.

GOAL 2 Evaluate if there is a link between the first and second experiments.



- EXPERIMENT I -

SHIFTING F1

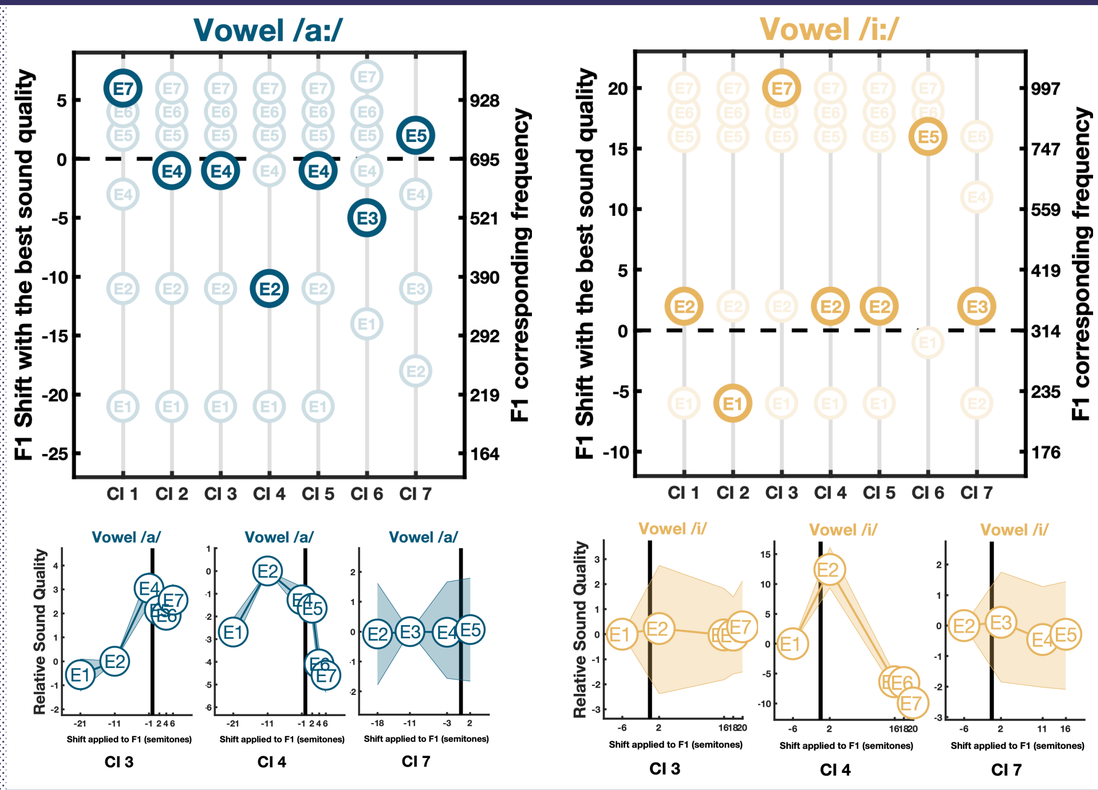
PAIRED COMPARISONS

BAYESIAN ANALYSIS

RANK SHIFTS & ELECTRODES

i | i

+5 +4 +3 +2 +1 = +1 +2 +3 +4 +5



Results from the F1 shifting experiment The top panels show the electrodes tested (associated with the shift and frequency value) for the different participants. The highlighted electrodes are those with the best sound quality rating. The bottom panels present three sets of results from individual participants, corresponding to the output of the Bayesian model. The participants' responses were chosen to highlight different response patterns.

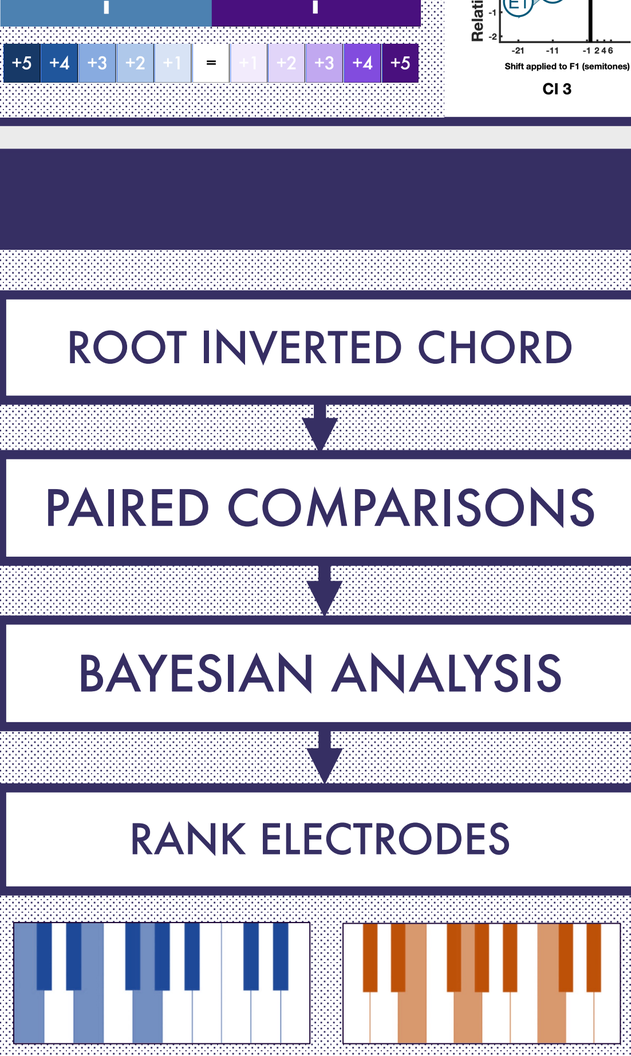
- EXPERIMENT II -

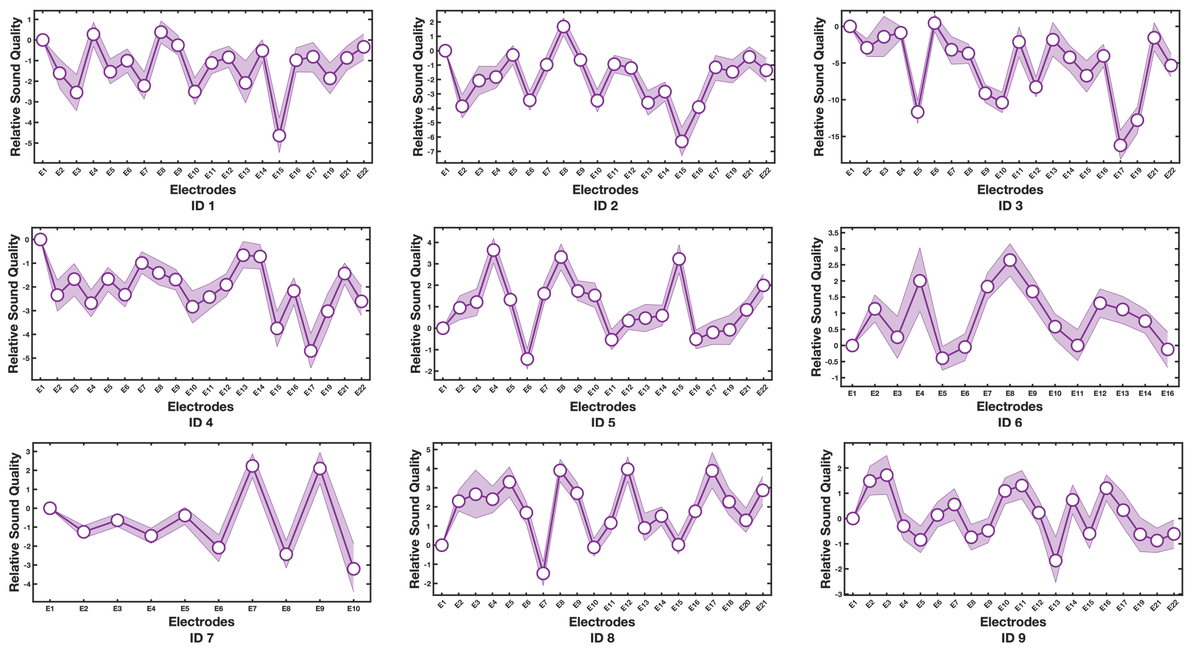
ROOT INVERTED CHORD

PAIRED COMPARISONS

BAYESIAN ANALYSIS

RANK ELECTRODES





Results from the Inverted Chord experiment. The response patterns of nine participants are presented here. At this stage, it is not possible to detect a common pattern of response among the different participants. Additionally, no link was found between the responses in this experiment and those in the F1 shifting experiment.

[1]FTPM or Frequency-to-place mismatch it refers to the difference between the acoustic frequency information conveyed by a given electrode and the frequency corresponding to the tonotopic location stimulated by that electrode.

[2]ENI or Electrode Neuron Interface quality it refers to the effectiveness of transmission between a given stimulating electrode and the neuronal interface it stimulates.

- Experiment I -

- Link between vowels is tricky
- Processing is limited to F1
- Sound quality is not necessarily relevant

- Experiment II -

- Pre-processing is too complex
- The question needs to be adjusted

